Unit 2 Classwork — Exponent Rules — Answer Key

Exponent Rules · Power/Zero/Negative Exponents

A. Exponent Rules — Product & Quotient

Problem 1 Score 3

Simplify: $x^4 \cdot x^3$

Answer: x^7

Problem 2 Score 3

Simplify: $y^8 \div y^3$

Answer: y^5

Problem 3 Score 4

Simplify: $(2x^3)(3x^2)$

Answer: $6x^5$

Problem 4 Score 4

Simplify: $15a^6 \div 3a^2$

Answer: $5a^4$

Problem 5 Score 4

Simplify: $a^2 \cdot a^5 \cdot a$

Answer: a^8

B. Power, Zero & Negative Exponents

Problem 1 Score 3

Simplify: $(x^3)^4$

Answer: x^{12}

Problem 2 Score 3

Simplify: 7^0

Answer: 1

Problem 3 Score 4

Simplify: $(2x^2)^3$

Answer: $8x^6$

Problem 4 Score 4

Simplify: $(5x^4)^0$

Answer: 1

Problem 5 Score 4

Simplify: 3^{-2} (write as a fraction)

Answer: $\frac{1}{9}$

Problem 6 Score 4

Simplify: x^{-4} (write as a fraction)

Answer: $\frac{1}{x^4}$

Problem 7 Score 5

Evaluate: -4^2 vs $(-4)^2$ — what is $(-4)^2$?

Answer: 16